the inner planets, including earth, and there will no longer even be a notion of a rising and setting sun, at least not here.

But these are modern scientific recognitions. To the ancients, cyclic phenomena seemed eternally cyclic. And to many, the cyclic phenomena, running their course and continuously returning to begin anew, were the primary phenomena. The cycles of days and seasons set the rhythm of work and life, so it is no wonder that some of the oldest recorded cosmologies envisioned the unfolding of the world as a cyclical process. Rather than positing a beginning, a middle, and an end, a cyclic cosmology imagines that the world changes through time much as the moon changes through phases: after it has passed through a complete sequence, conditions are ripe for everything to start afresh and initiate yet another cycle.

Since the discovery of general relativity, a number of cyclic cosmological models have been proposed; the best-known was developed in the 1930s by Richard Tolman of the California Institute of Technology. Tolman suggested that the observed expansion of the universe might slow down, someday stop, and then be followed by a period of contraction in which the universe got ever smaller. But instead of reaching a fiery finale in which it implodes on itself and comes to an end, the universe might, Tolman proposed, undergo a *bounce:* space might shrink down to some small size and then rebound, initiating a new cycle of expansion followed once again by contraction. A universe eternally repeating this cycle— expansion, contraction, bounce, expansion again—would elegantly avoid the thorny issues of origin: in such a scenario, the very concept of origin would be inapplicable since the universe always was and would always be.

But Tolman realized that looking back in time from today, the cycles could have repeated for a while, but not indefinitely. The reason is that during each cycle, the second law of thermodynamics dictates that entropy would, on average, rise.[8] And according to general relativity, the amount of entropy at the beginning of each new cycle determines how long that cycle will last. More entropy means a longer period of expansion before the outward motion grinds to a halt and the inward motion takes over. Each successive cycle would therefore last much longer than its predecessor; equivalently, earlier cycles would be shorter